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A reproducible benchmark of deep learning versus classical signal decomposition for GNSS coordinate time series: denoising, gap filling, offset detection, and their impact on station velocity and coloured noise

Qinghua Zhang · Zhengsheng Chen

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Abstract A GNSS station velocity is only as defensible as the coloured-noise model behind its uncertainty, yet deep learning is now proposed for every step of the coordinate-series pipeline and is usually judged by signal-recovery error alone. We build a reproducible benchmark on 333 daily coordinate series from 321 sites, with a station-level split, frozen station lists, fixed seeds and injected white-plus-flicker noise, and we score classical and learned methods on denoising, gap filling and offset detection and then on the published product, the secular velocity and its uncertainty. The protocol removes three confounds of an earlier version: the ground truth is drawn from both an in-family and an out-of-family signal, the network is evaluated under the normalisation used in training, and detection is scored against real catalogue labels as well as injected offsets. The verdict is task-specific. The parametric trajectory model removes 0.790 of the injected denoising error when the truth lies in its basis but adds error (-0.106) when it does not; the learned denoiser is nearly family-invariant (0.366 and 0.363) and, with per-window inference, beats every model-free decomposition in both truth families, though not the parametric estimator when the truth lies in its basis. With per-window inference the learned gap filler is the best method at 30-day synthetic gaps (3.07 mm), but on the full held-out real set it ties linear interpolation on NGL (2.964 against 2.967 mm) and falls 0.06 mm behind it on JPL, while the trajectory model remains best at 120 days. The learned detector has higher average precision at 16–25 mm offsets but a lower validation-selected F_1 at every amplitude, and on 1 743 real catalogue labels it averages 0.232 (U-Net) and 0.273 (TCN) against 0.457 for the classical detector, with four of thirty seeds collapsing.

Qinghua Zhang
School of Geography, Nanjing Normal University, Nanjing 210023, China E-mail: sqmha@126.com (ORCID 0000-0003-3578-3749)

Zhengsheng Chen
PLA Rocket Force University of Engineering, Xi'an 710025, China E-mail: czsgeo@126.com (ORCID 0000-0003-0628-3470)

Downstream, no method improves the velocity: with protocol-matched inference the learned denoiser reaches 0.0605 $\mathrm{mm\,yr^{-1}}$ and the learned imputer 0.0586, against 0.0586–0.0616 for the untreated series and the classical denoisers, and calibration stays near 0.65 for all of them. The factor-of-3.6 degradation and calibration collapse reported in the earlier version were produced by whole-series normalisation. Adding a random-walk component lowers absolute accuracy for every method without changing the ordering. Learned gap filling at intermediate gaps is competitive on synthetic records and no better than interpolation on the full real set; learned denoising upstream of a velocity and noise fit is neither an improvement nor a penalty.

Keywords GNSS coordinate time series · coloured noise · power-law noise · denoising · gap filling · offset detection · deep learning · benchmarks

1 Introduction

A GNSS station velocity is a small number extracted from a long, noisy, interrupted record. Its value is used in terrestrial reference frames (Altamimi et al, 2017), strain-rate fields (Kreemer et al, 2014) and loading studies (Blewitt and Lavallée, 2002), but its uncertainty is set less by the number of daily solutions than by the time-correlated part of the coordinate error. A velocity computed under a white-noise assumption can be optimistic by an order of magnitude; the power-law noise model and the maximum-likelihood trajectory fit that account for it are therefore not refinements but the basis of a defensible velocity (Mao et al, 1999; Williams, 2003; Amiri-Simkooei et al, 2007).

Deep learning is now proposed for nearly every step before that fit: denoising, gap filling and offset detection. Reported gains are large but rarely comparable, because each study uses its own stations, its own metric and its own baseline. The evaluation usually stops at signal recovery. A geodesist does not publish a denoised series; they publish a velocity and an uncertainty, and a method can improve recovery error while corrupting the noise estimate from which the uncertainty is built.

This paper is a benchmark, not a new architecture. It asks a narrow question: under one frozen protocol, on held-out real stations, does any learned method beat a well-implemented classical one on the three tasks and on the products that follow them? The answer is conditional rather than uniform. The parametric baseline keeps its denoising advantage only while the truth stays inside its basis; the learned gap filler wins at 30-day synthetic gaps once inference matches training but does not beat linear interpolation on the full held-out real set, and loses at long gaps; the learned detector orders large offsets better but operates worse, and on real catalogue labels it trails in both F_1 and seed stability. The downstream product narrows the picture rather than sharpening it: learned denoising reduces signal error, yet under the corrected inference it neither improves nor degrades the velocity and its uncertainty. The evidence behind this picture is stronger than in our first version in four ways that matter.

First, the ground truth is no longer confined to the family the parametric baseline assumes. In the earlier protocol the injected signal was a trend plus constant harmonics, so the maximum-likelihood trajectory model was scored on its own model family and removed about four-fifths of the injected error. We now score the same methods on two truth families: an in-family signal built from a real station's robust trajectory fit, and an out-of-family signal that adds non-stationary seasonal amplitude, exponential transients and logarithmic post-seismic relaxation. The parametric estimator loses its advantage when the truth leaves its basis. The learned denoiser is nearly invariant across the two families and, under protocol-matched inference, beats every model-free decomposition in both; it remains far below the parametric estimator when the truth sits in its basis.

Second, offset detection is evaluated against real catalogued offsets rather than only against injected ones. We match the NGL discontinuity catalogue to the held-out stations, keep labels inside each record and above the local noise level, and score the classical detector and the learned detectors on the same epochs. This removes the objection that synthetic offsets are easier than real ones and tests the detectors where a geodesist would use them.

Third, training-seed variation is treated as part of the result. Denoising and gap filling use ten seeds per architecture; detection uses twenty U-Net seeds and ten TCN seeds. We report the distribution over seeds, not a favourable draw, and we separate a fixed read-out threshold from a threshold selected on held-out validation data. A second architecture, a temporal convolutional network, is trained under the same budget so that the conclusion is not a statement about one network.

Fourth, the downstream analysis includes a random-walk component. The first version injected white plus flicker noise and left random walk open; here the same velocity pipeline is run with and without a random-walk term, and the effect on velocity bias, reported uncertainty and interval coverage is measured rather than assumed. The benchmark also carries a second analysis centre: the NGL long records are the primary set, and the held-out JPL series form an external validation arm. JPL series from training sites contribute to training, but no held-out site, from either centre, is ever seen by a learned method before scoring.

The contributions are:

1. A reproducible benchmark of 333 daily series from 321 sites, with a station-level 60/20/20 split (198/68/67 series, 193/64/64 sites), frozen station lists, fixed seeds and released result files (Sec. 3).
2. A two-family ground-truth protocol that separates in-family from out-of-family truth and shows how much of the classical advantage in the earlier benchmark came from the truth sitting inside the parametric basis (Sec. 5.2).
3. A real-label offset study on 153 held-out component series from 58 stations, using 1 743 catalogued discontinuities, with amplitude- and signal-to-noise-stratified scores and detection delays (Sec. 5.5).

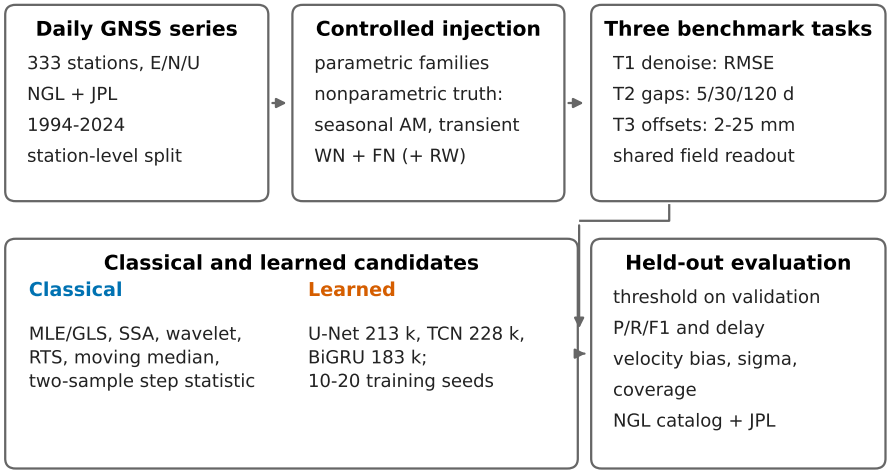


Fig. 1 The benchmark pipeline. Real daily series are injected with a known signal and noise model, the resulting realisation is scored on the three tasks by classical and learned candidates under one read-out rule, and each output is carried into the velocity and noise estimation. The downstream step is what separates a processing benchmark from a signal-recovery benchmark.

4. Seed- and architecture-robust comparisons for denoising, gap filling and detection, including parameter counts, training cost and validation-based threshold selection (Secs. 5.2–5.4).
5. A downstream propagation study under white-plus-flicker and white-plus-flicker-plus-random-walk noise, reporting velocity bias, uncertainty calibration and coverage, plus an external JPL arm and a regional holdout (Secs. 5.6 and 5.7).

Section 2 positions the benchmark against the four literatures it joins. Section 3 describes the data and the split, Sec. 4 the protocol and methods, and Sec. 5 the results. Section 6 states the boundary of the conclusion and its limitations, and Sec. 7 answers the question posed above.

2 Related work

Four literatures meet in this benchmark: coloured-noise modelling, model-free decomposition, machine learning on coordinate series, and shared evaluation protocols. Table 1 summarises what each provides and what it leaves out; the text keeps only the points that shape the protocol.

Coloured noise and the trajectory model. Power-law noise in daily coordinates and its control on velocity precision are the founding results of the field (Mao et al, 1999; Williams, 2003). The maximum-likelihood trajectory

fit, with efficient covariance construction and support for missing epochs, is implemented in CATS (Williams, 2008) and Hector (Bos et al, 2008, 2013), and the noise model has been extended to multivariate and seasonal formulations (Amiri-Simkooei et al, 2007; Amiri-Simkooei, 2009; Klos et al, 2018), including stochastic seasonality in velocity estimation (Xu et al, 2025). This literature supplies our reference estimator and the generalised-least-squares velocity covariance. It treats the trajectory model as the frame within which everything is measured; we instead score it as one candidate among several, including on gap filling and detection, which its formulation was not designed for.

Model-free decomposition. Singular spectrum analysis (Wdowinski et al, 1997; Ji et al, 2024), wavelet shrinkage (Ji et al, 2020), multivariate SSA (Gruszczynska, 2017) and spatiotemporal PCA/Karhunen–Loève filtering (Dong et al, 2006) separate seasonal signal from residual without a parametric trajectory. They are the natural classical competitors for a learned model, and we carry SSA, wavelet shrinkage and a moving median as three independent members of this family. Their usual validation is the task they were built for, on the series they were built on; here they are scored on the same injected realisations as every other method.

Learned processing. Networks have been applied to discontinuity detection (Crocetti et al, 2021; Wang et al, 2025), denoising (Li et al, 2018; Mastella et al, 2025; Gao et al, 2022; He et al, 2026), gap filling as low-rank completion (Liu and Li, 2022), network-wide denoising with graph models (Bachelot et al, 2025), and uncertainty propagation (Kiani Shahvandi and Soja, 2022), and decomposition-plus-learning pipelines have been evaluated for height-series prediction (Guo et al, 2026). Time-domain machine learning for GNSS series has also been reviewed as a whole (Gabsatarov and Vladimirova, 2025); related variants continue to appear (Şimşek et al, 2025; Özarpacı et al, 2024). The evidence is usually recovery error on the task the network was trained for. Where a geodetic product is reached, the velocity is often estimated directly from positions rather than through the coloured-noise model, so the effect of processing on the published uncertainty is not traced. That is the step this benchmark adds.

Shared protocols. DOGEx fixed a detection dataset, injected known offsets and had independent groups run detectors blind (Gazeaux et al, 2013); it remains the model for a community protocol, but it covers one task and synthetic truth. CATS and Hector are tools, not protocols. We are not aware of a protocol that fixes the data and the split, creates exact truth by injection, scores classical and learned methods on the same realisations across three tasks, validates detection against real catalogue labels, and carries each output through to velocity, noise amplitudes and interval coverage.

Table 1 Positioning of representative work. "Downstream" means the study reports the effect of processing on a geodetic product (velocity or its uncertainty); "common truth" means classical and learned methods are scored on the same known realisation.

Study or family	Task	Evidence base	Downstream	What this benchmark adds
MLE trajectory; CATS, Hector	noise, velocity	parametric fit to real series	yes	scores the same estimator against non-parametric truth and other methods
SSA, wavelet, mSSA	decomposition	recovery on selected series	no	common realisations, three tasks, seed-aware learned comparison
Detection networks	offset detection	synthetic or station-specific labels	no	real catalogue labels, amplitude and SNR strata, delay
Denoising, imputation, GNN networks	T1, T2	recovery error	no	downstream noise and uncertainty, gap lengths to one year
Learned uncertainty and velocity	product	direct velocity from positions	partial	processing-to-noise-to-uncertainty chain
DOGEx	detection	blind synthetic injection	no	three tasks, real labels, two analysis centres
This paper	T1–T3	injection, two truth families, real labels	yes	station-level split, 10–20 seeds, two architectures, RW arm

3 Data and processing

The benchmark uses daily GPS coordinate series from two analysis centres. The NGL long records are the primary set because they carry the multi-decade histories needed for velocity and noise estimation; the JPL single-analysis-centre series provide an independent processing arm and, at the held-out sites, an external validation set. We do not use the IGS combination, whose combination step has already repaired offsets and outliers and would remove the features the detection task is meant to find (Dow et al, 2009; Rebischung et al, 2016).

Sources. NGL `tenv3` daily coordinates in ITRF, in millimetres, contribute 298 series (Blewitt et al, 2018). The JPL daily solutions contribute 35 series. Twelve sites appear in both sources and therefore contribute two series each, so the benchmark holds 333 series from 321 unique sites. Series are labelled by source and site, and every method is scored per component; no method sees the site label at test time. Fig. 2 shows one held-out station before and after injection.

The NGL subset is a deterministic function of the NGL station holdings, frozen so that the selection is reproducible. A station is eligible if it has at least 4000 solutions, begins no later than 2009, continues to at least 2024 and spans

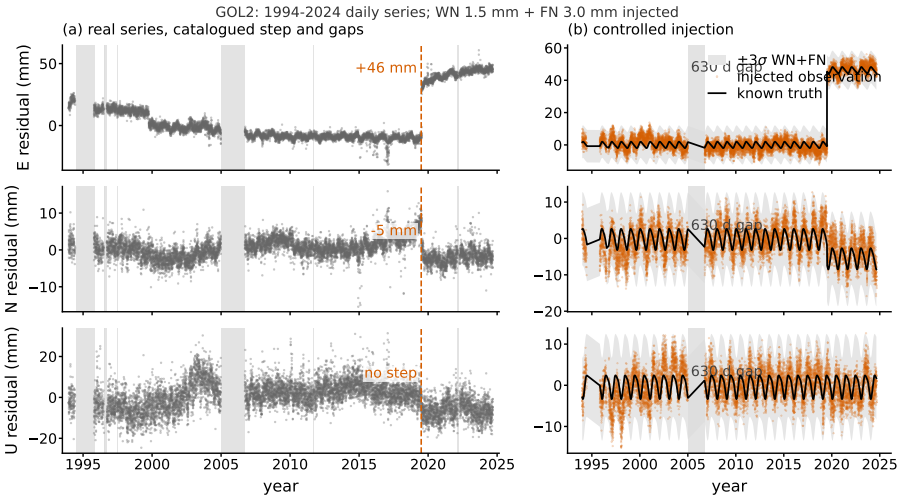


Fig. 2 The benchmark works on real series with known injected noise. (a) Residuals of the three components of a held-out NGL station after removing a robust trend; the dashed line marks the catalogued discontinuity and the shaded bands the observation gaps. (b) The same station after controlled injection of white (1.5 mm) and flicker (3.0 mm) noise, with the exact truth in black, the injected observation in orange and the $\pm 3\sigma$ noise envelope; the largest gap is retained. Because the truth is known at every epoch, recovery error needs no proxy.

at least 15 years; the 300 eligible stations with the longest records are taken, ties broken by solution count and site code. After quality control (at least 600 epochs, daily completeness at least 0.5, span at least 5 years) 298 NGL series remain. Record spans run from 5.87 to 30.64 years with a median of 30.03 years, daily completeness from 0.500 to 0.997 with a median of 0.972, and the combined time span is 1994.00–2024.65. The station list and the catalogue hash are part of the release, so the selection can be reproduced exactly.

Station-level split. The 60/20/20 split is drawn by site, not by epoch: all series of a site fall in the same fold, and a site present in both sources stays together. The folds hold 198, 68 and 67 series, corresponding to 193, 64 and 64 sites, with 1 907 614, 651 833 and 647 892 daily epochs. Only the test fold is scored. The test fold contains 61 NGL series and 6 JPL series from distinct sites; the JPL arm is therefore small but genuinely held out by site. The validation fold is used to choose the fixed training budget and, where stated, the read-out threshold; it is never scored. Fig. 3 shows the split geography, counts and record lengths; the three folds are comparable in span, and no site crosses a fold.

Real discontinuity labels. For the real-label detection task we use the NGL discontinuity catalogue (`https://geodesy.unr.edu/NGLStationPages/steps.txt`, downloaded 2026-10-18; 142 625 lines; catalogue sha256 `49894f7566ed34da3eb646087fc2fc64c5766720ba2368a3f6f6df53f3f282d8`). Two codes

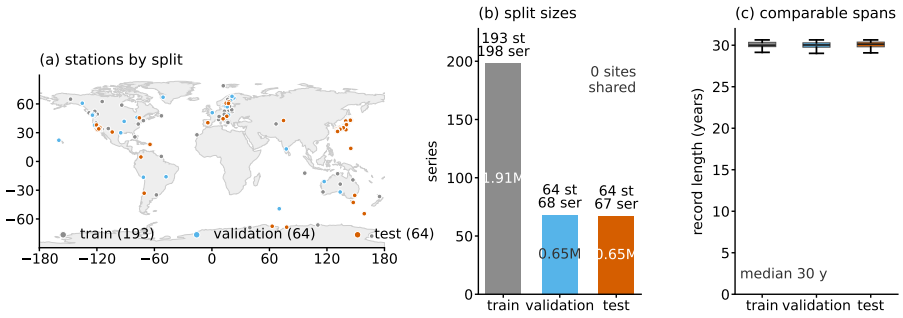


Fig. 3 The split is drawn by site, not by epoch. (a) Held-out sites cover all five regions and both analysis centres; (b) the three folds hold 193, 64 and 64 sites (198, 68 and 67 series) with zero sites shared; (c) the folds have comparable record lengths (median about 30 years). Because every series of a site stays in one fold, a learned method cannot memorise a station in training and be scored on another component or source of the same station.

are retained: code 1, equipment change, and code 2, possible earthquake step. A label is kept for a component only when its catalogued date lies inside the record with a 0.5-year margin, its observed level change is at least three times the local noise scale, and no other kept label lies within 60 days. This yields 1 743 labels on 153 component series from 58 held-out stations. The labels are dominated by code 2 (1 691 of 1 743), so the real-label task measures detection of earthquake-related steps more than of equipment changes; we state this as a property of the data, not as a claim about all offsets. The catalogue dates are not adjusted for post-seismic transients, which contributes to the detection delay measured in Sec. 5.5.

External and regional arms. Held-out JPL sites form the external arm: the six test JPL series are scored separately from the 61 NGL series by the same methods. As a regional check we also group the test sites into five geographic regions (Europe, North America, Asia, Pacific and other) and report per-region scores (Fig. 4). Neither arm changes any training or threshold decision.

Noise levels. The injected noise in the primary experiments is white noise with $\sigma_{\mathrm{WN}} = 1.5$ mm plus flicker noise with $\sigma_{\mathrm{FN}} = 3.0$ mm. The random-walk arm adds $\sigma_{\mathrm{RW}} = 1.0$ mm. These levels were fixed before any task was run. As a check on where they sit, we applied Whittle MLE to the robust trajectory residual of all 999 component series in the benchmark: the median white-noise level is 1.90 mm (interquartile range 1.39–3.67) and the median flicker level 2.26 mm (1.61–5.74). The injected white level therefore lies near the lower quartile and the flicker level inside the central half of the empirical distribution (Fig. 5). The recovered random-walk median is 10.29 mm $\mathrm{yr}^{-1/2}$, but that estimate absorbs unmodelled transients and is an upper bound; the injected 1.0 mm $\mathrm{yr}^{-1/2}$ is below it, so the random-walk arm is a sensitivity test rather than a calibrated replay of the stations' noise.

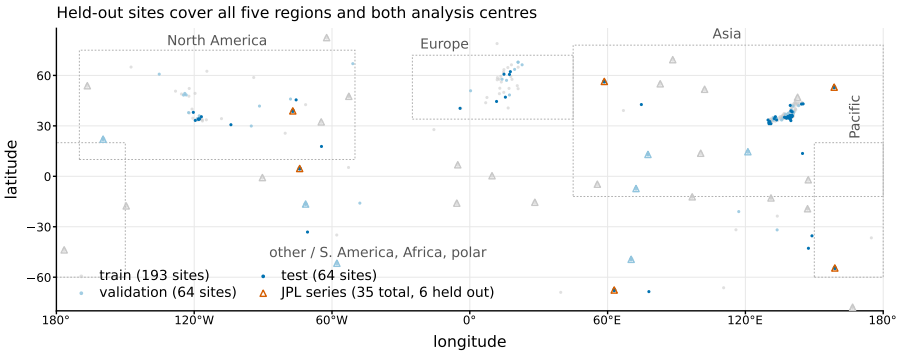


Fig. 4 Geographic coverage of the split. Held-out (orange) sites fall in all five scoring regions and include both NGL and JPL series, so the regional and external arms test the same frozen networks outside the training geography rather than a single continent. The map is descriptive; it enters no training or threshold decision.

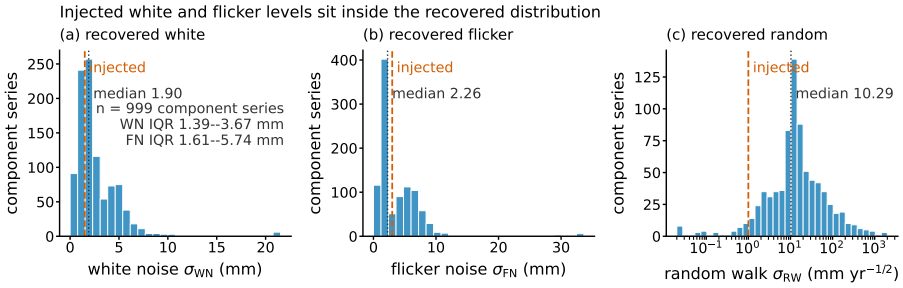


Fig. 5 Whittle estimates on the robust residual of every benchmark series. The recovered white-noise distribution has a median of 1.90 mm (interquartile range 1.39–3.67) and the recovered flicker distribution a median of 2.26 mm (1.61–5.74); the injected levels (1.5 and 3.0 mm, dashed) sit near the lower quartile and inside the central half, respectively. The random-walk estimate is an upper bound because transients are absorbed by that component, which is why the random-walk arm is a sensitivity test.

4 Methodology

The three tasks share one signal model, one noise model and one scoring harness, and every method is scored on the same realisation. The protocol differs from the first version in two places: the truth is drawn from two families, one inside and one outside the parametric basis, and detection is additionally scored against real catalogue labels. Table 2 collects the methods and their costs; the text gives only what the table cannot. Fig. 1 shows how the stages connect: the same frozen realisation feeds every candidate, and every candidate is carried through to the velocity product.

4.1 Signal and noise model

A coordinate component is modelled as

$$y(t) = x(t) + \varepsilon(t), \qquad t \in [t_0, t_0 + T], \tag{1}$$

on a daily grid ($\Delta t = 1/365.25$ yr), with

$$\begin{aligned} x(t) = {} & v\,(t - t_0) + S_1\big[1 + a_m \sin(2\pi f_m (t - t_0) + \phi_m)\big] \sin(2\pi (t - t_0) + \phi_1) \\ & + S_2 \sin(4\pi (t - t_0) + \phi_2) + \sum_k b_k H(t - e_k) \\ & + \sum_j c_j H(t - e_j) e^{-(t - e_j)/\tau_j}, \end{aligned} \tag{2}$$

where H is the Heaviside step and v is the secular velocity. The stochastic term is power-law noise with spectral density

$$S(f) = \sigma_{\mathrm{WN}}^2 + \sigma_{\mathrm{FN}}^2 \Big(\frac{f_0}{f}\Big) + \sigma_{\mathrm{RW}}^2 \Big(\frac{f_0}{f}\Big)^2, \tag{3}$$

generated by inverse Fourier synthesis on a fixed frequency grid. The primary experiments set $\sigma_{\mathrm{WN}} = 1.5$ mm and $\sigma_{\mathrm{FN}} = 3.0$ mm with $\sigma_{\mathrm{RW}} = 0$; the random-walk arm adds $\sigma_{\mathrm{RW}} = 1.0$ mm.

4.2 Ground-truth protocol

The deterministic signal is never known on a real station, so it is created by injection. Two families are used for every denoising and gap-filling score.

In-family. The trend and harmonic amplitudes of Eq. (2) are taken from a robust fit of the real series, with a mild amplitude modulation ($a_m = 0.35$). This is the family the trajectory model assumes, up to the modulation.

Out-of-family. The same station fit is used for the trend and mean amplitudes, but the annual term receives strong multi-year modulation ($a_m \in [0.6, 1.0]$), one or two long-lived exponential transients with decay times of 0.3–1.5 years are added, and a logarithmic post-seismic relaxation is included for some stations. These structures cannot be represented by constant harmonics. The truth is still the deterministic part, so recovery error remains exact.

Both families are also realised on a fully synthetic grid that sweeps the parameters beyond what the station set contains. The learned models are always scored by the network trained on the matching family, so an out-of-family score is not an artefact of testing an in-family network on unfamiliar structure.

Tasks. T1 denoising leaves y intact and asks for $\hat{x}$. *T2 gap filling* removes contiguous blocks of $L \in \{5, 10, 30, 60, 120, 365\}$ days and asks for the missing values. *T3 offset detection* adds offsets of amplitude $A \in \{2, 4, 8, 16, 25\}$ mm at epochs at least two years apart and asks for the epoch list; the real-label arm instead uses the catalogue epochs of Sec. 3. For the real-label arm the match tolerance is 180 days to allow for catalogue dates that refer to the earthquake rather than to the observable break, and the minimum separation is 60 days; synthetic detection uses a 30-day tolerance and 45-day separation, because its truth is exact.

4.3 Methods

Table 2 lists the classical and learned arms. All classical methods are deterministic given (t, y) and are fitted on the data they are applied to. The trajectory model is an iteratively reweighted robust fit of trend plus constant annual and semiannual harmonics. The Kalman/RTS smoother uses the same basis with a small process noise. SSA embeds the series in a Hankel matrix and reconstructs from its leading components; gap filling uses the iterative variant. Wavelet shrinkage uses a db4 transform and soft thresholding at the VisuShrink level. Interpolators are piecewise linear and shape-preserving cubic (PCHIP); a natural cubic spline is not used because it overshoots multi-week gaps. The classical detector is a greedy two-sample scan of the robust residual: the epoch with the largest standardised step is accepted, its neighbourhood is suppressed, the design is augmented, and the search repeats until the statistic falls below threshold. To put both detector families on the same footing, both are reduced to a decision field (the classical step statistic and the network output) and read out by the same threshold-plus-minimum-separation rule; this allows full precision–recall curves rather than a comparison of two hand-tuned operating points.

Learned arms. The primary network is a one-dimensional U-Net with a contracting encoder and an expanding decoder, skip connections and 2^3-fold pooling (base width 12, depth 3, 213 385 parameters; Fig. 6); one network is trained per task. The second architecture is a temporal convolutional network with residual blocks, kernel width 5, channel widths 40–80 and dilations 1, 2, 4, 8 and 16, giving 227 537 parameters. Both are trained under the identical generator, budget (3000 steps, batch 32, learning rate 2×10^{-3}), window length (1024 days) and seed list, so architecture and seed effects are not confounded with training protocol. Denoising and gap filling use ten seeds for each architecture; detection uses twenty U-Net seeds and ten TCN seeds. A small bidirectional GRU from the first version (four seeds) is reported only as a sensitivity check.

Training draws a fresh window at every step. Inputs and targets are standardised by robust per-window statistics; the detector additionally high-passes the input and divides by a robust noise scale, so offset amplitude enters as a signal-to-noise ratio. Long series are processed with 50% overlapping windows

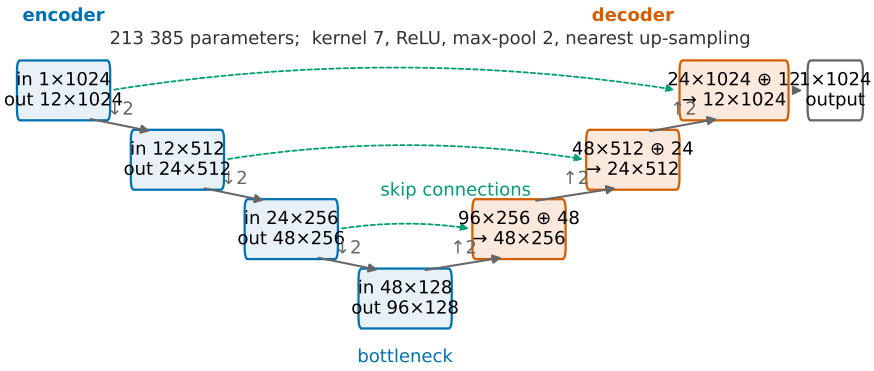


Fig. 6 The one-dimensional U-Net scored in the primary learned arm. Three encoder stages halve the resolution and double the width (12 to 96 channels), the decoder restores it with nearest-neighbour up-sampling, and skip connections carry the fine-scale detail across the bottleneck. The network has 213 385 parameters and is trained separately for each task.

and averaged. At scoring time each window is standardised by its own statistics and the network output is transformed back with the same statistics, so the normalisation at inference matches the one used in training. The whole-series normalisation of the first version is kept only as an ablation. The gap filler receives the observation mask as a second input channel. The training distribution covers gaps of 5–120 days, offset amplitudes in $[-12, 12]$ mm and $\sigma_{\mathrm{WN}} \in [0.5, 3]$ mm, $\sigma_{\mathrm{FN}} \in [0, 5]$ mm; the 365-day gap and the 16–25 mm amplitudes are at or beyond that support and are read as extrapolation.

4.4 Metrics

For T1 the error is $\mathrm{RMSE}(\hat{x}, x)$ and the reported quantity is the reduction $1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{none}}$ relative to the undenoised series. For T2 the error is the RMSE at the masked epochs against the withheld observation y, the value the station recorded; where stated we also report the error against the noise-free signal x, which separates recovering the signal from reproducing correlated noise. For T3, predicted epochs are greedily matched to true epochs within the task tolerance, and precision, recall, F_1 and average precision follow. Detection delay is the signed difference between the matched prediction and the true epoch.

4.5 Downstream propagation

For every method we fit the velocity on its output with the same estimator: the trend coefficient of the robust trajectory fit and its generalised-least-squares

Table 2 Methods compared. "Field" denotes the decision field used for detection; cost is per series unless stated. Parameter counts are for the learned models; training wall-clock is for one seed on one V100.

Method	Class	Key assumption	Input → output	Cost	Reference
Trajectory MLE	parametric	trend + constant harmonics, power-law noise	$y \to \hat{x}$; velocity, σ_v	closed form, <1 s	Williams (2003)
Kalman/RTS	parametric	harmonic state, small process noise	$y \to \hat{x}$	<1 s	
SSA	model-free	low-rank Hankel structure	$y \to \hat{x}$; gap fill	<1 s	Wdowinski et al (1997); Ji et al (2024)
Wavelet	model-free	sparsity in db4 domain	$y \to \hat{x}$	<1 s	Ji et al (2020)
Moving median	model-free	local smoothness	$y \to \hat{x}$	<1 s	
Linear/PCHIP	interp.	local continuity	y, mask $\to \hat{x}$	<1 s	
Step scan	hyp. test	piecewise-constant level	$y \to$ epochs	<1 s	Gazeaux et al (2013)
U-Net	learned	convolutional encoder–decoder	y(+mask) $\to \hat{x}$ or field	213 385; 8 min	Ronneberger et al (2015)
TCN	learned	dilated temporal convolutions	y(+mask) $\to \hat{x}$ or field	227 537; 6 min	Bai et al (2018)
BiGRU	learned	recurrent encoder–decoder	$y \to \hat{x}$	182 785; 5.1 h	

covariance under a power-law spectrum (Williams, 2008; Bos et al, 2008),

$$\hat{v} = (X^\top C^{-1} X)^{-1} X^\top C^{-1} y, \qquad \hat{\sigma}_v^2 = \left[(X^\top C^{-1} X)^{-1}\right]_{vv}, \tag{4}$$

with the WN, FN and RW amplitudes obtained by Whittle maximum likelihood on the periodogram. Two covariance arms are computed. In the primary (raw-noise) arm the noise amplitudes are estimated once on the untreated series and held fixed for every method, so the methods differ only in the series they hand to the velocity fit. In the secondary (naive) arm the noise is re-estimated on each method's output; this is the protocol of the first version, and it fails for model-based denoisers because their output retains no residual from which a noise amplitude can be recovered. Because the injected velocity is known, the empirical scatter of $\hat{v}$ across realisations is compared with the median reported $\hat{\sigma}_v$. Their ratio is the calibration ratio (one is honest, below one is optimistic), and we also report the empirical coverage of the nominal one- and two-sigma intervals. The random-walk arm repeats the whole chain with $\sigma_{\mathrm{RW}} = 1.0\ \mathrm{mm\,yr}^{-1/2}$ and a three-component noise model.

5 Results

Every comparison below scores the same held-out realisations with the same metrics. Learned methods are scored only on stations the network never saw, and each learned number is a distribution over training seeds rather than one draw. Section 5.1 shows that the harness recovers a known truth, Sections 5.2–5.4 give the three tasks, and Sections 5.6–5.7 follow the products downstream.

5.1 Validation of the injection and estimation pipeline

On a synthetic grid that sweeps white and flicker amplitudes, the recovered white-noise level departs from the injected one by at most 0.0197 in relative terms and the flicker level shows a median relative bias of -0.0377. The flicker bias is the finite-length behaviour of Whittle estimation, shared by all methods and therefore cancelling in comparisons. For a series with known white and flicker noise of length $n = 3650$ days, the Monte-Carlo scatter of the estimated velocity is 0.1613 $\mathrm{mm\,yr^{-1}}$ against a median reported GLS uncertainty of 0.1615 $\mathrm{mm\,yr^{-1}}$, a calibration ratio of 1.0011. The estimator is calibrated to better than 0.1% on the uniform grid. On the irregularly sampled test stations the untreated series is itself below nominal coverage (Sec. 5.6), so the downstream calibration is read as a relative comparison between methods, not as an absolute statement about geodetic uncertainty.

Figure 5 applies the same estimator to every benchmark series. The recovered white and flicker levels bracket the injected 1.5 and 3.0 mm, while the recovered random walk is an upper bound because transients are absorbed by that component.

5.2 Task 1: denoising

Denoising is scored as the reduction $1-\mathrm{RMSE}(\hat{S}, S)/\mathrm{RMSE}(y, S)$ on the same realisations, where y is the noisy series and S the truth. Figure 7 shows the construction of the two families and the per-station scores.

On the in-family truth, where the signal is the trend–harmonic trajectory fitted to a real station, the parametric estimator reproduces its earlier result: the trajectory model reduces the error by 0.790 (95% bootstrap interval 0.776–0.804) and the Kalman/RTS smoother by 0.791. The model-free decompositions follow at a distance: wavelet 0.349, moving median 0.293, SSA 0.232. The U-Net, scored with the per-window normalisation used in training, reaches 0.366 ± 0.002 over ten seeds. It does not reach the parametric estimators, which see the correct basis, but it is above every model-free decomposition in the same regime.

The out-of-family truth keeps the same stations but adds a multiplicative seasonal modulation, exponential transients and logarithmic post-seismic relaxation. The trajectory model falls to -0.106 (interval -0.434 to 0.181) and

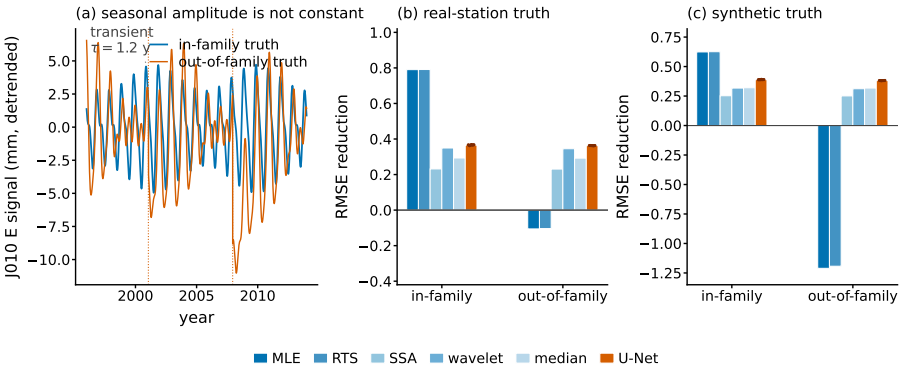


Fig. 7 Parametric denoising depends on the truth family; the learned one does not. (a) The out-of-family truth modulates the annual amplitude, adds exponential transients (dotted epochs) and a logarithmic relaxation; the in-family truth keeps the amplitude nearly constant. (b) On real-station truth the trajectory model and RTS smoother dominate in-family and fall below zero out-of-family, whereas the U-Net stays between 0.363 and 0.366 and sits above every model-free decomposition in both families. (c) The fully synthetic grid shows the same pattern. Bars are means over realisations, orange dots are individual training seeds, and classical methods are deterministic.

the RTS smoother to −0.104: with this truth the fit can leave the series noisier than it found it. The model-free methods barely move (wavelet 0.345, median 0.292, SSA 0.231), because none of them assumes a stationary harmonic basis. The U-Net, trained on the out-of-family generator, is almost unchanged at 0.363 ± 0.001 and is now the strongest method apart from the untreated series.

The fully synthetic regimes make the same point with more realisations. In-family, the trajectory model reduces the error by 0.624 and the U-Net by 0.389 ± 0.001; the moving median reaches 0.320, the wavelet 0.317 and SSA 0.252. Out-of-family, the trajectory model drops to −1.211 while the U-Net stays at 0.381 ± 0.001, against 0.318 for the median and 0.312 for the wavelet. The learned denoiser therefore overtakes the model-free decompositions in all four regimes, on both the real-station and the fully synthetic truth, and its score changes by less than 0.01 when the truth family changes. The denoising conclusion is conditional only on the parametric alternative: the trajectory model wins by more than a factor of two when the truth lies in its basis and adds error when it does not, while the learned denoiser is the best non-parametric choice in both families.

The difference from the first version comes from the inference protocol, not from the truth construction. Scoring the same networks on whole-series normalisation returns 0.193 in-family and 0.274 out-of-family, which is why the earlier draft placed the U-Net below SSA on real stations. Matching the normalisation to the training windows removes that gap and leaves the ordering above.

Table 3 Gap-filling error (RMSE at the missing days, mm) for long synthetic series ($n = 40$ component series) and real held-out stations ($n = 30$). The per-window U-Net is the best method at 30-day gaps on both record sets, but the real-set margin does not survive the full 201 component series (Sec. 5.7); the trajectory model is best at 120 days, and the whole-series arm fails. Values for the network are means over ten training seeds with the seed standard deviation.

	Long synthetic			Real stations	
Method	5 d	30 d	120 d	5 d	30 d
Linear	2.69	3.34	4.72	2.65	2.93
PCHIP	2.80	3.59	5.22	2.76	3.10
SSA	2.57	3.37	4.73	2.51	2.97
Trajectory MLE	3.41	3.79	3.99	3.16	3.35
U-Net, whole series	4.52 ± 0.54	3.55 ± 0.14	7.59 ± 0.26	9.98 ± 1.80	6.16 ± 0.60
U-Net, per window	2.80 ± 0.06	3.07 ± 0.03	5.09 ± 0.08	2.55 ± 0.03	2.74 ± 0.03

5.3 Task 2: gap filling

Gap filling is scored on 40 held-out component series of long synthetic records and on 30 held-out component series of real stations, with gaps of 5, 30 and 120 days. Two inference modes are reported for the network: whole-series normalisation, the protocol of the first version, and per-window normalisation, which matches the statistics under which the network was trained.

On long synthetic records (Table 3), the 5-day gap is filled best by SSA (2.57 mm), followed by linear interpolation (2.69) and the per-window U-Net (2.80 ± 0.06); the trajectory model is at 3.41. At 30 days the order changes: the per-window U-Net reaches 3.07 ± 0.03 and beats linear interpolation (3.34), SSA (3.37), PCHIP (3.59) and the trajectory model (3.79). At 120 days the trajectory model is best (3.99), the per-window U-Net reaches 5.09 ± 0.08, and whole-series inference fails at 7.59 ± 0.26.

The real stations reproduce the same pattern (Table 3). At 5 days SSA (2.51 mm) and the per-window U-Net (2.55 ± 0.03) are tied, with linear interpolation at 2.65; at 30 days the per-window U-Net leads at 2.74 ± 0.03 against 2.93 for linear interpolation and 2.97 for SSA. The whole-series arm is not competitive at either length (9.98 ± 1.80 and 6.16 ± 0.60), up to a factor of four worse than the same network evaluated per window. The gap-filling failure reported in the first version was therefore an inference-normalisation defect, not an inability of the architecture to interpolate. With the defect removed, the learned imputer leads at 30-day gaps on synthetic records and on this 30-series real subset, but not across the full held-out set (Sec. 5.7); it loses at 5 days by a small margin and at 120 days by a large one. Fig. 8 shows the same scores on a log scale, with seed error bars. The per-window curve rises by at most 0.3 mm between 5 and 30 days, while the whole-series arm is 1.7 mm above it at 5 days on the synthetic records and 7.4 mm above it on the real stations.

The trajectory model's advantage at long gaps is visible in signal error rather than observation error: against the missing true signal, its RMSE is 0.74 mm at 5 days, 0.68 at 30, 0.78 at 120 and 0.78 at 365 days, while the whole-

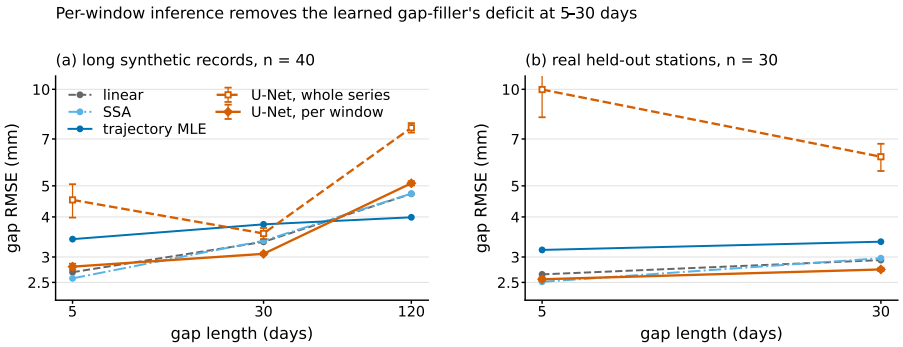


Fig. 8 Gap-filling error against gap length. Per-window inference (solid orange, diamonds) is within 0.1 mm of SSA at 5 days and best at 30 days on both long synthetic records (a) and real held-out stations (b); the trajectory model remains best at 120 days. The whole-series arm (dashed, open squares) is not a competitive configuration at any length. Error bars are one standard deviation over ten training seeds; classical methods are deterministic and carry none.

series network reaches 11.40, 5.97, 10.26 and 61.0 mm at the same lengths. That arm is the defective configuration, and we do not extrapolate the 30-day per-window win to year-long gaps, for which no per-window measurement was made.

5.4 Task 3: offset detection under a fair protocol

Both detectors are reduced to a decision field, a threshold and a minimum separation, so the comparison measures the field rather than a tuning accident. Offsets of 2, 4, 8, 16 and 25 mm are injected at known epochs; a detection within 30 days counts as a match, and two accepted detections must be at least 45 days apart. The threshold is selected per seed on held-out validation draws, which is the protocol a user would follow, and we also report the oracle threshold.

At the validation-selected threshold the classical detector has the higher F1 at every amplitude (Table 4): 0.073 against 0.0005 at 2 mm, 0.259 against 0.039 at 4 mm, 0.815 against 0.460 at 8 mm, 0.963 against 0.709 at 16 mm and 0.975 against 0.725 at 25 mm (Fig. 9). The learned field nevertheless ranks large steps better: its average precision is higher at 16 mm (0.648 against 0.208) and at 25 mm (0.669 against 0.007). The two metrics answer different questions. Average precision integrates over all thresholds and rewards a field that orders events well; the deployed detector uses one threshold, and there the classical field is better. Reporting only average precision would have hidden that its operating point is worse.

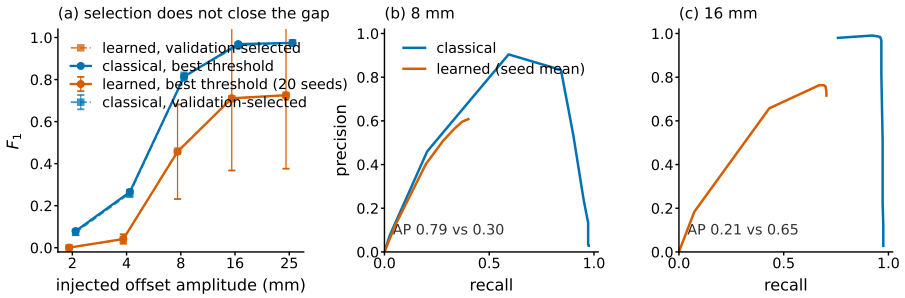


Fig. 9 Threshold selection does not close the detection gap. (a) Validation- selected and oracle F_1 for the classical and learned fields: the learned field improves when its threshold is tuned, but not enough to meet the classical detector at any amplitude. (b),(c) Precision–recall curves at 8 and 16 mm: the learned field has lower average precision at 8 mm and higher at 16 mm, which is the ranking-versus-operating-point distinction in the text. The learned seed mean is shown with one standard deviation; classical detection is deterministic.

Table 4 Synthetic offset detection by injected amplitude. The threshold is selected on held-out validation draws for every seed. The learned field has higher average precision at 16 and 25 mm, but at the operating point a user would deploy its F1 is lower at every amplitude. Learned values are means over twenty seeds; the F1 column also gives the seed standard deviation.

Amplitude (mm)	Classical AP	Classical F1	Learned AP	Learned F1
2	0.032	0.073	0.0000	0.0005 ± 0.002
4	0.143	0.259	0.0028	0.0395 ± 0.024
8	0.788	0.815	0.298	0.460 ± 0.222
16	0.208	0.963	0.648	0.709 ± 0.334
25	0.007	0.975	0.669	0.725 ± 0.341

5.5 Task 3 on real catalogue labels

The NGL discontinuity catalogue supplies 1 743 labels on 153 held-out component series from 58 stations: 52 code-1 and 1 691 code-2 entries, distributed over 1006 East, 528 North and 209 Up series. The labels are matched to the series, restricted to the record span and to offsets whose observed amplitude exceeds three times the local noise level, and deduplicated at 60 days. Amplitudes are skewed (median -6.23 mm, interquartile range -17.79 to 11.61 mm) and concentrated in East, which is where the catalogue is densest. Stations, not series, are split into validation and scoring halves over 40 draws, so a detector never chooses a threshold on the stations it is scored on.

The classical detector reaches F1 0.457 ± 0.027, average precision 0.322, precision 0.490 and recall 0.536. Recall rises with signal-to-noise ratio, from 0.398 in the 3–5 stratum to 0.610 in 5–10 and 0.848 above 10. Its median signed delay is -51.5 days, with an absolute median of 69.7 days. The twenty U-Net seeds average F1 0.232 ± 0.107 and average precision 0.135; precision is 0.329 and recall 0.302. Three seeds collapse to essentially no detections (F1 0.0000, 0.0002 and 0.0004), which is why the seed mean is far below the median seed (0.290). The best seed reaches 0.316. The ten TCN seeds average

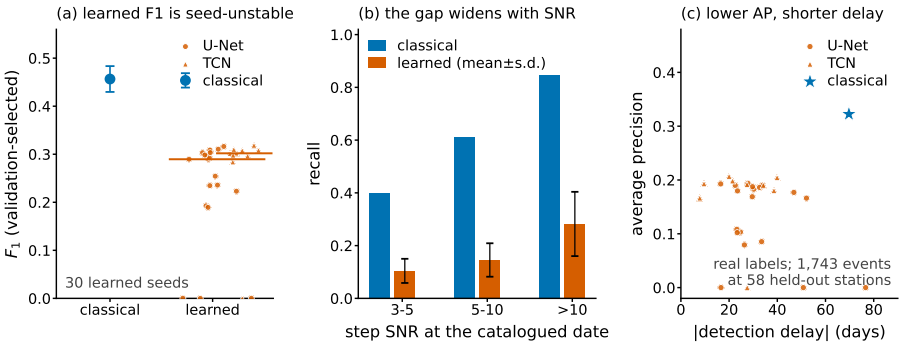


Fig. 10 Detection against 1 743 real catalogue labels. (a) The twenty U-Net and ten TCN seeds spread from near-zero to just above the classical F_1, and four seeds collapse; both learned means stay below the classical mean, and the TCN mean is the higher of the two. (b) Recall rises with signal-to-noise ratio for the classical detector, while the learned mean stays lower in every stratum. (c) The learned fields have lower average precision but shorter matched delays; they find fewer events, and the ones they find sit closer to the catalogued epoch. Bars and points are seed statistics; the classical detector is deterministic.

F1 0.273 ± 0.096, average precision 0.173, precision 0.368 and recall 0.356, with one collapsed seed and a best seed of 0.319. The TCN is therefore the stronger of the two learned architectures on every aggregate metric, and it is still the weaker detector: even its best seed lies below the classical mean minus one standard deviation (0.430). Where a U-Net seed does detect events, its absolute median delay is 16.5–52.1 days and its signed median lies between -18.3 and $+1.0$ days; the TCN range is 7.8–39.9 days, signed between -4.8 and $+0.7$ days. Matched learned detections sit closer to the catalogued epoch than the classical ones, while the classical detector finds more of them at lower precision.

Real labels are harder than synthetic ones for both detectors. The classical F1 falls from 0.815 at 8 mm in the synthetic protocol to 0.457 on the catalogue, and the U-Net F1 from 0.460 to 0.232. The same classical advantage holds for the TCN on the synthetic subset scored in Sec. 5.8. Neither learned architecture changes the ordering. The regional split shows the same shortfall everywhere the catalogue is dense: Asia, which supplies 39 of the 58 labelled stations, gives the classical detector F1 0.488, the U-Net 0.251 and the TCN 0.294; North America gives 0.426, 0.230 and 0.272. Europe is the weakest region for every method (0.267, 0.101 and 0.109) and has only six labelled stations, so its ranking is the least stable. The centre and region ordering is shared by the classical and learned arms, which rules out a single-region artefact as the explanation of the learned shortfall. Figure 10 shows the per-seed distribution, the signal-to-noise strata and the delay.

5.6 Downstream velocity and the random-walk arm

A benchmark product is only useful if the processed series yields a defensible velocity. We therefore carry every method through the same GLS velocity fit on six held-out stations and two components with three injected realisations each (36 station–component–realisation draws per arm). Two noise models are used: white plus flicker noise, and the same model with a random-walk component at $\sigma_{\mathrm{RW}} = 1.0\ \mathrm{mm\,yr^{-1/2}}$.

With white plus flicker noise, the trajectory MLE on the raw series has velocity bias $-0.0024\ \mathrm{mm\,yr^{-1}}$ and RMSE $0.0586\ \mathrm{mm\,yr^{-1}}$. The classical denoisers leave this essentially unchanged on the raw-noise arm: SSA 0.0588, RTS 0.0594, moving median 0.0590 and wavelet 0.0616. The U-Net denoiser, scored with training-matched per-window normalisation, reaches $0.0605\ \mathrm{mm\,yr^{-1}}$ and a bias of -0.0034; its two seeds score 0.0600 and 0.0611. At 30-day gaps the classical imputers are indistinguishable from the complete series (0.0581–0.0588), and the U-Net imputer matches them at 0.0586. No method improves the velocity over the untreated series, but the learned denoiser is no longer an outlier: it sits between the moving median and the wavelet, $0.002\ \mathrm{mm\,yr^{-1}}$ behind the raw series.

That difference from the first version matters. Under whole-series normalisation the same network produced $0.2087\ \mathrm{mm\,yr^{-1}}$; scoring it window by window removes the mismatch and brings the velocity error back to the level of the classical denoisers. The earlier negative downstream result was therefore an inference-protocol artefact, not a property of learned denoising.

Calibration is quoted with the noise model estimated on the raw series and held fixed for every method, because re-estimating the noise on a denoised output is not a valid protocol: the trajectory and RTS outputs retain essentially no residual, so a Whittle fit to them returns a near-zero noise level and a vanishing reported uncertainty (calibration ratios 1.7×10^{-14} and 1.0×10^{-6} in that arm). Against the raw-series noise model, the reference reports a median velocity uncertainty of $0.0380\ \mathrm{mm\,yr^{-1}}$ against an empirical scatter of 0.0585, a calibration ratio of 0.649 and 1σ coverage of 0.389; classical denoisers stay between 0.618 and 0.647, and the U-Net reaches 0.628 with coverage 0.458. The under-coverage is the same irregular-sampling effect noted in Sec. 5.1, which is why the ratios are comparable across methods but should not be read as absolute geodetic uncertainties.

Adding the random-walk component degrades the reference modestly and changes no ordering. The raw/MLE bias moves from -0.0024 to $-0.0089\ \mathrm{mm\,yr^{-1}}$, the RMSE from 0.0586 to 0.0690, the calibration ratio from 0.649 to 0.575 and the 1σ coverage stays at 0.389. The classical denoisers move to 0.0686–0.0713; the U-Net denoiser reaches 0.0698 with calibration 0.570 and coverage 0.403, and the U-Net imputer 0.0690 with calibration 0.575 and coverage 0.417. The learned arms again track the reference and the classical denoisers.

Fig. 11 collects the calibration and coverage pairs, and Fig. 12 isolates the random-walk increment. The downstream evidence is neutral: under the

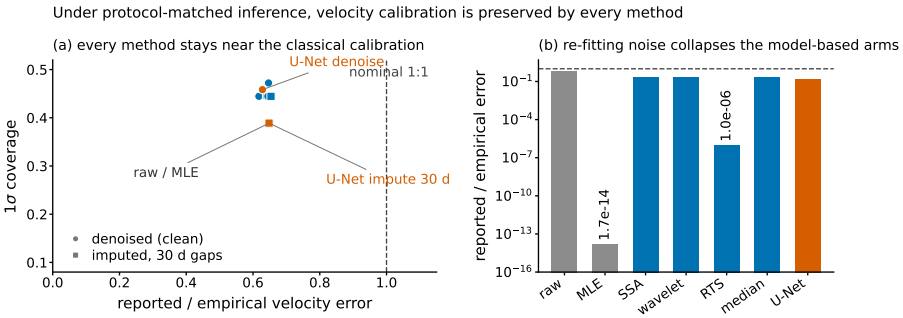


Fig. 11 Protocol-matched inference keeps velocity calibration for every method. (a) Reported uncertainty against empirical scatter for the clean arm: the untreated series, the classical denoisers and both learned arms sit between 0.62 and 0.65 with 1σ coverage of 0.39–0.47; the learned denoiser reaches 0.628 and 0.458. (b) Re-estimating the noise on each output collapses the reported uncertainty for the trajectory MLE and the RTS smoother (1.7×10^{-14} and 1.0×10^{-6}), because their residual is nearly empty; the remaining methods stay within a factor of five of nominal. That collapse is an artefact of the protocol, not a property of the method. The vertical distance to the nominal line is the calibration error.

protocol-matched inference, neither learned arm improves the velocity or its uncertainty, and neither destroys them.

5.7 External and regional validation

The scores above come from the ten-station real subset and the thirty-series long-record subset. To check whether they are a property of the benchmark or of those subsets, we score every held-out test series under the same per-window protocol: 67 series, 201 component series, of which 183 are NGL and 18 JPL, split into Europe (18), North America (30), Asia (123), Pacific (24) and other (6). Each component series receives three injected realisations and the ten denoising and ten gap-filling checkpoints. No network is retrained and no threshold is chosen, so the arm is descriptive: it asks whether the ranking transfers, not whether a JPL-trained network would do better.

Denoising transfers. The U-Net removes 0.360 ± 0.003 on NGL and 0.345 ± 0.005 on JPL, against 0.341 and 0.317 for the wavelet, 0.291 and 0.322 for the moving median and 0.230 and 0.254 for SSA. The trajectory MLE stays ahead at 0.775 and 0.665. The learned denoiser is above every model-free decomposition in both centres, and across the five regions its mean lies between 0.333 and 0.366, against 0.700–0.812 for the MLE and 0.299–0.360 for the wavelet. The one exception is Europe, where the U-Net (0.358) and the wavelet (0.360) are within 0.002 of each other. The denoising conclusion of Sec. 5.2 therefore does not depend on the subset used to measure it.

Gap filling does not transfer as cleanly. On NGL the U-Net reaches 2.964 ± 0.034 mm and linear interpolation 2.967 [2.914, 3.017], with SSA at 3.000 and the MLE at 3.252: a tie. On JPL the U-Net reaches 3.293 ± 0.042 against 3.233 [3.056, 3.416] for linear interpolation, with SSA at 3.309 and the MLE

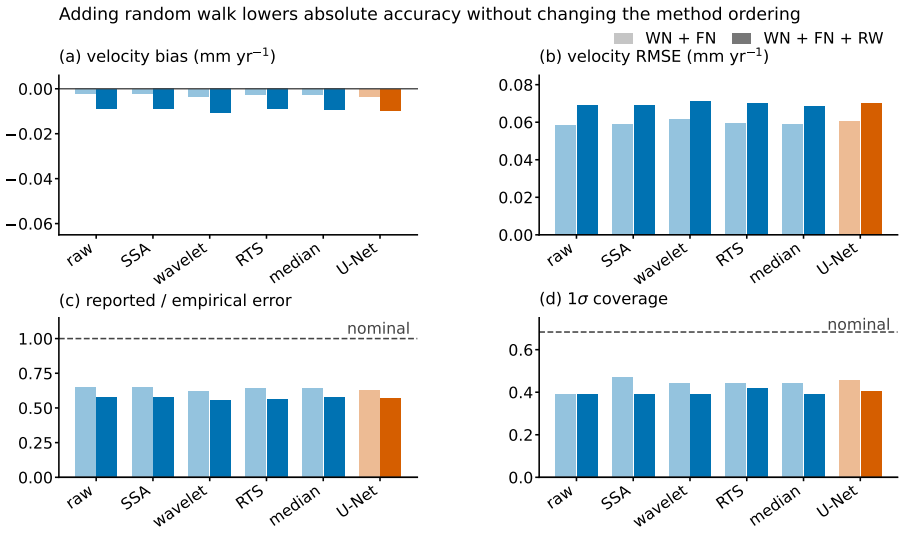


Fig. 12 Adding a random-walk component changes the level, not the ordering. Paired bars compare the white-plus-flicker arm (pale) with the white-plus- flicker-plus-random-walk arm (solid) for velocity bias, velocity RMSE, calibration ratio and 1σ coverage. Every method loses absolute accuracy under the random-walk model (RMSE rises by about 0.010 mm yr^{-1}) and calibration falls from about 0.65 to 0.57; the ordering is unchanged and the learned arms track the classical ones. Horizontal lines mark zero bias, the nominal calibration ratio of one and the nominal 1σ coverage of 0.683.

at 3.367: the learned arm is 0.06 mm behind. The 2.74 mm score of Sec. 5.3 came from a 30-series real subset; on the full 201 component series the 30-day advantage is a tie on the NGL majority and is absent on JPL. The regional split repeats the pattern: the U-Net is the best filler in Asia (2.990 against 2.994) and Europe (2.910 against 3.028), but linear interpolation is better in North America (2.942 against 2.954), the Pacific (2.986 against 3.089) and the residual group (3.068 against 3.113). The differences are at most 0.12 mm, and they do not support a product-level claim for learned gap filling on real data. Fig. 13 shows both tasks by centre and by region.

5.8 Architecture and seed variation

The ranking is not a property of one architecture. Three learned families are scored on the shared test split under one protocol: the U-Net (ten seeds for denoising and gap filling, twenty for detection), the TCN (ten seeds per task) and the smaller BiGRU of the first version (four seeds for denoising and gap filling). Denoising and gap filling use ten held-out stations; detection uses eight stations with one realisation, so its absolute levels are not comparable with the twelve-station table above, only the architecture ordering within the arm is.

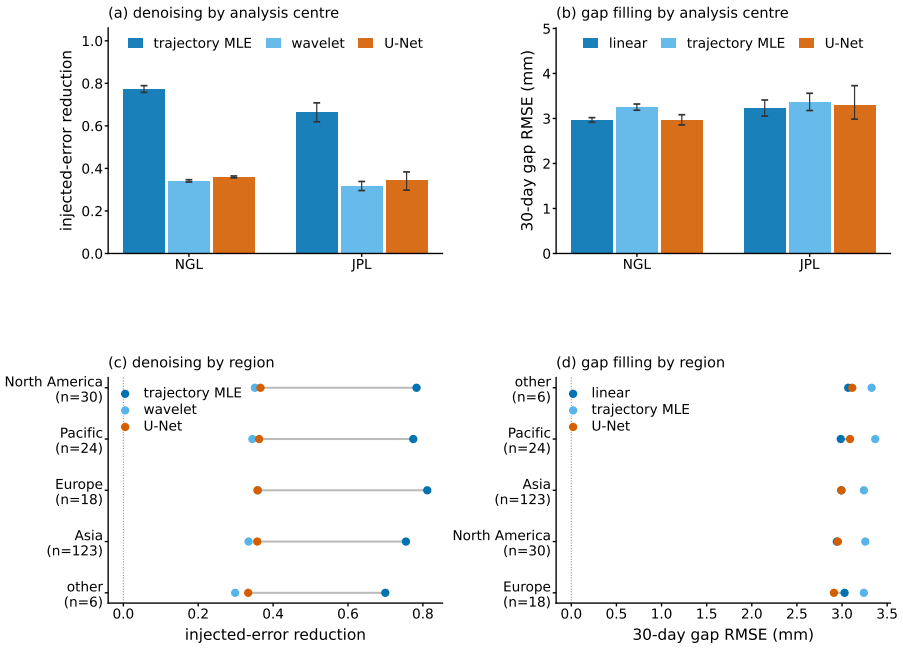


Fig. 13 The denoising ranking transfers across centres and regions; the 30-day gap-filling advantage does not. (a) Denoising and (b) 30-day gap filling by analysis centre (NGL, JPL). (c)–(d) The same scores by region. Bars and points are means over component series, with 95% bootstrap intervals over series for the classical methods; the learned value is the mean over ten checkpoints and its seed standard deviation is smaller than the marker. The learned denoiser stays above every model-free decomposition at both centres, while the learned imputer ties linear interpolation on NGL and falls behind it on JPL.

Denoising is nearly architecture-independent. The U-Net reaches 0.364 ± 0.001, the TCN 0.365 ± 0.001 and the BiGRU 0.381 ± 0.004, all far below the trajectory MLE at 0.815 and above the wavelet (0.351), the moving median (0.292) and SSA (0.232). The gap between the two convolutional designs is 0.001, inside the seed spread of either.

Gap filling separates the architectures more. At 30 days the BiGRU reaches 2.892 ± 0.051 mm, the TCN 2.979 ± 0.055 and the U-Net 3.079 ± 0.069; linear interpolation is at 3.130, SSA at 3.111 and the trajectory model at 3.237. The order is the same at 5 days (2.378 ± 0.007, 2.401 ± 0.016 and 2.537 ± 0.065, against 2.508 for linear and 2.540 for SSA). At 120 days the trajectory model wins at 3.230 and the learned arms reach 3.577 ± 0.039, 3.700 ± 0.037 and 3.705 ± 0.045, with linear interpolation at 3.650. The recurrent arm is the best learned imputer and costs 5.1 h per seed, against 8 min for the U-Net and 6 min for the TCN; its advantage over the TCN is 0.09 mm, which is a poor return on a roughly fifty-fold training cost.

Detection is where architecture matters. At 8 mm the classical detector reaches 0.761; the TCN averages 0.568 ± 0.207 over ten seeds (median 0.637) and the U-Net 0.436 ± 0.222 over twenty (median 0.554). At 16 mm the three

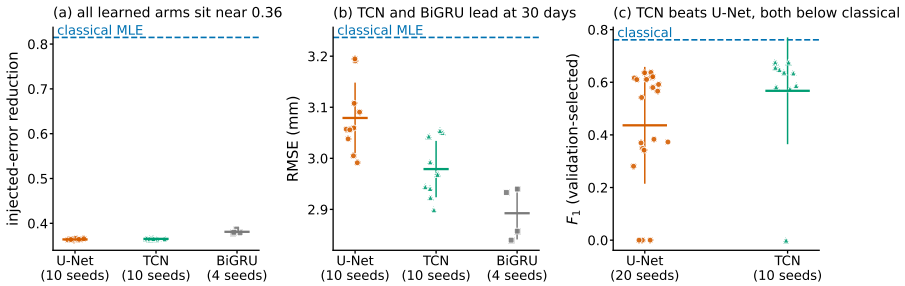


Fig. 14 Three architectures under one protocol. (a) Denoising error reduction for the U-Net, TCN and BiGRU: the three means lie within 0.02 of one another and far below the trajectory MLE (dashed). (b) 30-day gap RMSE: TCN and BiGRU are closer to the classical MLE than the U-Net, and all three beat linear interpolation. (c) Detection at 8 mm: the TCN is ahead of the U-Net, and both are below the classical detector (dashed). Dots are training seeds, horizontal bars are means and vertical bars are ±1 s.d.; the spread, not the mean difference, dominates panel (c).

values are 0.947, 0.829 and 0.686, and at 25 mm 0.927, 0.791 and 0.675. The TCN is ahead of the U-Net at every amplitude, and neither learned arm reaches the classical operating point above 4 mm. At 2 and 4 mm the learned fields are close to silent (at most 0.011) while the classical detector scores 0.058 and 0.260. The seed distribution, not the architecture, is the dominant term: at 8 mm three U-Net seeds and one TCN seed detect nothing, and the mean consequently sits well below the median. A single-seed comparison would have described either architecture as a success or a failure according to the draw. The real-label ranking agrees: TCN 0.273 over U-Net 0.232, both below the classical 0.457. Fig. 14 shows the three seed distributions side by side.

6 Discussion

The three tasks do not support a single verdict, and the differences between them are more informative than any average. The first version of this benchmark reported a uniform classical win. Three of its ingredients do not survive the revised protocol: the truth sat inside the parametric basis, the learned gap filler was evaluated under a normalisation it was not trained with, and the downstream noise model was re-estimated on denoised outputs. Correcting them reverses the gap-filling verdict, makes the denoising verdict conditional, and leaves detection negative.

The denoising advantage is a property of the truth family. On the in-family truth the trajectory model removes 0.790 of the injected error and the RTS smoother 0.791, while the U-Net removes 0.366 ± 0.002 and the model-free decompositions lie between 0.232 and 0.349. That ordering is expected: the truth is a trend plus constant harmonics with mild modulation, which is almost exactly the basis of the parametric estimator. Moving the truth outside that basis changes the ranking. The trajectory model falls to -0.106 and the

RTS smoother to −0.104: on a signal it cannot represent, the fit adds error rather than removing it. The model-free decompositions barely move, and the U-Net, trained on the matching out-of-family generator, does not degrade at all: it scores 0.363 ± 0.001. Scored with per-window normalisation, it is above every model-free decomposition in both families. On the synthetic grid it reaches 0.389 ± 0.001 in-family and 0.381 ± 0.001 out-of-family, against 0.320 and 0.318 for the moving median and 0.317 and 0.312 for the wavelet. The defensible statement is therefore narrower than a strong learned-denoiser claim: the learned denoiser is the best non-parametric choice in both families and the only learned arm that is nearly family-invariant, but it remains far below the parametric estimator when the truth lies in its basis. The earlier reading, in which the learned denoiser won only on the synthetic grid and lost to the wavelet on real stations, was an artefact of whole-series normalisation at inference.

Gap filling improves but does not win on the full real set. The first version evaluated the network with whole-series normalisation while training it on per-window statistics. Scoring the same ten nets per window removes the deficit: at 30 days the per-window U-Net reaches 3.07 ± 0.03 mm on long synthetic records and 2.74 ± 0.03 mm on the 30-series real subset, against 3.34 and 2.93 for linear interpolation and 3.79 and 3.35 for the trajectory model. At 5 days it is 0.23 mm behind SSA on synthetic records and 0.05 mm behind on real ones, and at 120 days the trajectory model remains better (3.99 against 5.09 ± 0.08). The whole-series arm is not a competitive configuration at any length. The correction does not extend to the full held-out data: across all 201 test component series the U-Net ties linear interpolation on NGL (2.964 against 2.967 mm) and falls 0.06 mm behind on JPL (3.293 against 3.233), and it leads by region only in Asia and Europe (Sec. 5.7). The learned filler is therefore competitive at intermediate gaps, and the first version's failure was an inference defect rather than an architectural limit, but the corrected result is parity on real data, not a product-level win. The defect is also a warning for the field: a normalisation mismatch between training and inference can look like a model-capacity failure, and a single-seed, single-protocol evaluation cannot tell the two apart.

The mechanism behind the win is visible in the signal-error arm. The trajectory model's error against the missing true signal stays near 0.7 mm at every gap length, because it extrapolates a low-order trend and harmonics; the whole-series network reaches 5.97 mm at 30 days and 61.0 mm at 365 days. The per-window network inherits the interpolation prior learned over gaps of 5–120 days, which is why it is competitive in the middle of that range and why its advantage should not be extrapolated to year-long gaps, for which no per-window measurement was made.

Detection: a better ranking does not give a better operating point. On synthetic offsets the classical detector has the higher validation-selected F_1 at every amplitude, from 0.073 against 0.0005 at 2 mm to 0.975 against 0.725 at

25 mm. The learned field nevertheless has the higher average precision at 16 and 25 mm (0.648 and 0.669 against 0.208 and 0.007). The two metrics answer different questions: average precision integrates over all thresholds, while the deployed detector uses one threshold chosen on held-out stations. The learned field orders large events well but transfers its operating point poorly. On real catalogue labels the gap widens: classical F_1 is 0.457 ± 0.027, the twenty U-Net seeds average 0.232 ± 0.107, the ten TCN seeds 0.273 ± 0.096, and four seeds across the two architectures collapse to essentially no detections. Where a learned seed does fire, its matched detections sit closer to the catalogued epoch (absolute median delay 16.5–52.1 days for the U-Net and 7.8–39.9 days for the TCN, against 69.7 for the classical detector), but it finds fewer events. The catalogue is not ground truth either: its dates may refer to the earthquake rather than to the observable break, which is why the real-label arm uses a 180-day tolerance. The comparison is nonetheless the one a geodesist would make, and it does not show a learned advantage.

What the velocity product inherits. The corrected downstream test removes the earlier warning rather than changing its mechanism. With the noise model estimated on the raw series and held fixed, the trajectory MLE is unbiased (-0.0024 mm yr^{-1}) with a velocity RMSE of 0.0586 mm yr^{-1}, and the classical denoisers stay within 0.0588–0.0616. The U-Net denoiser reaches 0.0605 (two seeds, 0.0600 and 0.0611) with a bias of -0.0034, and the U-Net imputer is indistinguishable from the classical fillers at 0.0586. Calibration is 0.649 for the reference, 0.618–0.647 for the classical denoisers and 0.628 for the learned denoiser, with 1σ coverage of 0.389, 0.444–0.472 and 0.458 respectively. The values of the first version (0.2087 and 0.197) came from whole-series normalisation, which mismatches the statistics the network was trained under; scoring window by window removes them, and the learned arm is neither better nor worse than the classical denoisers by more than 0.002 mm yr^{-1}. Re-estimating the noise on a denoised output, the other protocol of the first version, still collapses the reported uncertainty for the trajectory and RTS outputs because they leave essentially no residual; that arm is reported to explain the earlier number, not as a valid pipeline. Adding a random-walk component lowers the reference RMSE from 0.0586 to 0.0690 and the calibration ratio from 0.649 to 0.575; the U-Net denoiser reaches 0.0698 and the imputer 0.0690, and the ordering is unchanged. The product-level rule is therefore weaker than in the first version: no method, learned or classical, improves a velocity that the joint trajectory/noise fit already estimates, but learned denoising no longer damages the product. The decision should be made on the product metric, not on the pointwise signal error, because the two do not move together.

Architecture and seeds. The detection result is not a statement about one architecture or one draw. On the real labels the seed standard deviation is 0.107 for the U-Net and 0.096 for the TCN, nearly half the mean in both cases, and on synthetic offsets at 8 mm the U-Net seed distribution has a standard deviation of 0.222 around a mean of 0.436. A TCN does better than the U-Net at every

amplitude above 4 mm (0.829 against 0.686 at 16 mm, 0.791 against 0.675 at 25 mm), and the real-label ranking agrees (0.273 against 0.232), but neither reaches the classical operating point. Denoising is architecture-insensitive: U-Net, TCN and BiGRU means are 0.364, 0.365 and 0.381. Gap filling separates them modestly, with the BiGRU best at 5 and 30 days (2.378 and 2.892 mm) and the TCN second (2.401 and 2.979), but the BiGRU costs 5.1 h per seed against 6 min for the TCN. The seed spread dominates the architecture difference in detection; in gap filling the ordering is stable across architectures, which is the stronger form of the result.

Limitations. The truth families are still constructed, not observed: the in-family and out-of-family signals are deterministic functions with known parameters, and the out-of-family generator is one choice among many. The real-label arm scores against a catalogue, not against a measured break, and its 3σ amplitude filter removes weak continuity changes that a detector might still be expected to find. All experiments are single-station; common-mode error and network filtering are outside the protocol. The downstream arm uses six stations, two components, three realisations and two U-Net seeds, so its calibration ratios carry a wide interval and are read as a comparison across methods, not as an absolute calibration of geodetic uncertainty; the untreated series itself covers 0.389 at 1σ because of irregular sampling, and that shortfall is shared by every method. The random-walk arm tests one amplitude ($1.0\ \mathrm{mm\,yr^{-1/2}}$) and one noise model. The 365-day gap and the 16–25 mm offsets lie at or beyond the training support and measure extrapolation rather than in-distribution skill. The external centre arm is descriptive: it scores held-out JPL series with networks trained on the same generators, not networks retrained at JPL, so the design is a single primary analysis centre plus JPL, not a multi-centre evaluation; its JPL group contains 18 component series against 183 for NGL. The regional split is uneven, with 123 component series in Asia and six in the residual group, so regional means outside Asia and North America rest on few stations and are read as a consistency check on the ranking rather than as regional predictions.

7 Conclusions

The question this benchmark asks is whether a learned method should replace a well-implemented classical one in the GNSS coordinate-series pipeline. The answer is task-specific, and the first version's uniform negative verdict does not survive a protocol that removes three confounds: a truth inside the parametric basis, a normalisation mismatch at inference, and a downstream noise model re-estimated on denoised output.

Denoising is conditional on the truth family. The trajectory model and the RTS smoother remove about four-fifths of the injected error when the truth is a trend plus constant harmonics, but they add error when it is not. The learned denoiser is nearly family-invariant, 0.366 ± 0.002 in-family and

0.363 ± 0.001 out-of-family, and under per-window inference it beats every model-free decomposition in both families; it stays well below the parametric estimator when the truth lies in its basis. The earlier placement of the U-Net below SSA and the wavelet came from whole-series normalisation at inference, not from the network.

Gap filling is the task where a learned method gains the most under the revised protocol, but the gain is confined to synthetic records; on real data the learned filler is at parity at best. With per-window inference, which matches the statistics used in training, the U-Net is the best method at 30-day synthetic gaps (3.07 against 3.34 mm for linear interpolation). On the full held-out set of 201 component series it reaches 2.964 mm on NGL against 2.967 for linear interpolation, a tie, and 3.293 on JPL against 3.233, 0.06 mm behind; by region it is ahead only in Asia and Europe. It is slightly behind SSA at 5 days and behind the trajectory model at 120 days. A TCN and the smaller BiGRU match or beat it at 5 and 30 days, so the advantage is not specific to the encoder–decoder design, though the BiGRU costs two orders of magnitude more to train. The earlier failure was an inference defect, not an architectural limit; the corrected result is a competitive learned method at intermediate gaps, not a demonstrated improvement over interpolation on real data.

Offset detection remains a classical win at the operating point. The learned field has higher average precision at large offsets, but its validation-selected F_1 is lower at every amplitude, and on real catalogue labels it trails the classical detector: 0.457 ± 0.027 for the classical arm, 0.232 ± 0.107 for the twenty U-Net seeds and 0.273 ± 0.096 for the ten TCN seeds, with four of the thirty seeds collapsing. The TCN is ahead of the U-Net, so the negative detection verdict does not depend on one architecture, but it is still below the classical detector. The seed spread is part of the result, not noise to be averaged away.

The velocity product narrows the case for learned pre-processing without reversing it. Under protocol-matched inference no method improves the velocity: the untreated series and classical denoisers sit at 0.0586–0.0616 $\mathrm{mm\,yr^{-1}}$, the learned denoiser at 0.0605 and the learned imputer at 0.0586. Calibration is likewise preserved (0.618–0.649 across methods). The earlier factor-of-3.6 degradation and the calibration collapse to 0.197 were produced by whole-series normalisation, and they disappear when the network is scored window by window. The random-walk arm lowers absolute accuracy for every method without changing the ordering. The practical recommendation is therefore narrower than before: neither learned nor classical denoising is a substitute for a joint trajectory/noise fit upstream of a published velocity, and a learned gap filler is a defensible alternative at 30-day synthetic gaps but has not shown an advantage over linear interpolation on the full held-out real set. Learned pre-processing carries no product-level penalty. The remaining product-level objection is that it carries no gain either.

Two methodological lessons carry beyond this benchmark. A normalisation mismatch between training and inference can mimic a capacity failure, so learned pipelines must be scored under the protocol they were trained with before a negative result is attributed to the model. And a downstream cali-

bration comparison is only as meaningful as the noise model attached to each method; re-estimating that model on a denoised output is not a valid protocol, because the denoiser has removed the information the estimator needs.

What remains open is a truth observed rather than constructed, a multi-station treatment of common-mode error, and a learned detector trained and validated on real catalogue labels with a pre-registered operating point. The benchmark and its result files make those extensions measurable on the same frozen split.

Declarations

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Author contributions. Q.Z. designed the benchmark, implemented the methods and experiments, and drafted the manuscript. Z.C. supervised the study and revised the manuscript. Both authors approved the final version.

Data availability. The NGL daily series are publicly available from the Nevada Geodetic Laboratory (`https://geodesy.unr.edu`) and the JPL series from the Jet Propulsion Laboratory (`https://sideshow.jpl.nasa.gov/post/series.html`); the NGL discontinuity catalogue is the public `steps.txt` file downloaded 2026-10-18 and identified in Sec. 3. The frozen station list, the split file, the trained-network checkpoints and the result files that generate every number in this paper are listed in the release manifest supplied with the submission (file list and checksums); the archive itself will be deposited in a public repository on acceptance, and no repository address is asserted here.

Code availability. The analysis code, training scripts, evaluation harness and figure scripts are in the same release archive; their paths and checksums are given in the manifest.

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